

A Bilayer LiF/C₆₀ Interfacial Strategy for High Efficiency Inverted PbS QD Photodetectors

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Abstract—This study proposes a bilayer interface engineering strategy that significantly enhances device performance in inverted PbS quantum dot photodetectors through the incorporation of an ultrathin LiF layer and a conventional C₆₀ layer. The LiF/C₆₀ bilayer structure effectively passivates surface traps, improves film flatness, and ensures favorable energy level alignment and efficient electron extraction. Experimental results demonstrate that devices employing this strategy exhibit nearly an order of magnitude reduction in dark current, an increase in external quantum efficiency from 15.45% to 28.90%, and a specific detectivity exceeding 8.47×10^{11} Jones—representing a more than sixfold improvement compared to unmodified devices.

Index Terms—Quantum dot, photodetector, bilayer interfacial engineering, light extraction, external quantum efficiency.

I. INTRODUCTION

COLLOIDAL PbS quantum dots (QDs) have emerged as attractive candidates for shortwave infrared (SWIR)

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photodetectors due to their bandgap tunability, solution-process compatibility, and potential for low-cost fabrication [1], [3]. In recent years, regular-structure QD photodetectors (typically built on an electron transport layer (ETL)/active layer/hole transport layer (HTL) stack) have demonstrated impressive conversion efficiency and well-controlled dark current density [4]. However, these devices often rely on glass substrates with ITO bottom electrodes, which are not compatible with Au electrodes in CMOS substrates [5].

In conventional regular-structure photodetectors, the active layer is typically formed by a one-step PbS QD ink onto an ETL, followed by solid-state ligand exchange to form a thin PbS-EDT layer as the HTL. This homojunction design, where both layers share the same kind of QD material, ensures intimate interfacial contact and suppresses leakage and recombination [6]. In contrast, inverted architecture requires the deposition of an ETL on top of the PbS ink, often using polymers or nanoparticles (NPs) such as PCBM and ZnO NPs. The resulting heterojunction interface tends to suffer from poor contact and increased traps [7]. Several efforts have been made to improve this interface, including solvent engineering, insertion of buffer layers [8], and the use of self-assembled monolayers [9]. However, these methods often suffer from limited process compatibility or insufficient defect suppression, result in limited efficiency.

In this work, we present a bilayer interfacial engineering strategy comprising an ultrathin LiF layer and a traditional C₆₀ layer. The LiF layer [10] effectively passivates surface traps and smooths the PbS film, while the C₆₀ layer ensures favorable energy level alignment and efficient electron extraction. Together, the LiF/C₆₀ bilayer minimizes recombination losses, suppresses dark current, and maintains film integrity. Devices fabricated with this bilayer interfacial strategy exhibit over a 8-fold increase in specific detectivity compared to unmodified controls, demonstrating the effectiveness of this CMOS-compatible interface design for high-performance inverted PbS QD SWIR photodetectors.

II. FABRICATION OF PbS QD PHOTODIODE

The PbS quantum dots were firstly synthesized via a hot injection method [11]. These oleic acid-capped PbS QDs (10 mg/mL in octane) were mixed with a lead halide solution (PbI₂ and PbBr₂ in DMF), resulting in phase transfer into DMF. After washing and centrifugation, the QDs were redispersed in a mixed n-butylamine/DMF solvent (4:1 v/v) to yield an PbS QD ink with high concentration of ~380 mg/mL.

The photodetector architecture consists of ITO/NiO_x/PbS-EDT/PbS-ink/LiF/C₆₀/PCBM/BCP/Ag as shown in Fig. 1a.

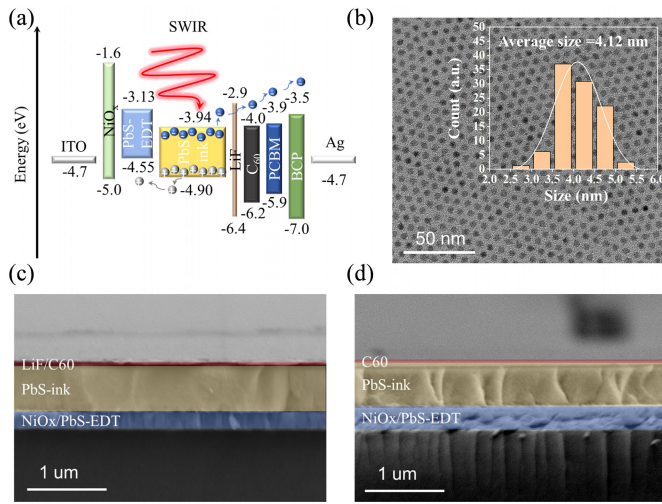


Fig. 1. (a) Energy level diagram of the PbS QD PD. (b) TEM image of PbS QDs. Cross-sectional SEM of devices with (c) and without (d) LiF modification layer.

ITO substrates were cleaned and treated with oxygen plasma, followed by spin-coating of NiO_x NPs at 2000 rpm and annealing at 150 °C. The PbS-EDT layer was fabricated via layer-by-layer solid-state ligand exchange, forming a ~50 nm film. The PbS ink (~400 nm) was spin-coated and annealed at 85 °C. A 1 nm LiF layer was deposited by thermal evaporation under vacuum, followed by 10 nm C₆₀. The LiF/C₆₀ bilayer were designed based on their well-studied functions of trap passivation, [12] energy band alignment, [13] and electron extraction [14] at the QD/ETL interface. PCBM and BCP were sequentially spin-coated, and a 100 nm Ag electrode was thermally evaporated to complete the device [15]. The active area of the device is 0.045 cm².

III. RESULTS AND DISCUSSION

The 1 nm LiF layer acts as a defect-passivating interfacial barrier that suppresses dark-state leakage while allowing efficient tunneling of photogenerated carriers under illumination. The transmission electron microscopy (TEM) image of PbS QDs (Fig. 1b) confirms a uniform size distribution with an average diameter of 4.12 nm, indicating high monodispersity. Cross-sectional SEM images of devices without (Fig. 1c) and with (Fig. 1d) LiF modification reveal clearly stacked multilayer structures. In both devices, the PbS-EDT and PbS ink layers dominate the vertical thickness, while the ETL is relatively thin and difficult to resolve. However, in the LiF-modified device, the PbS QD-ink layer appears smoother and more compact, with visibly fewer internal cracks or voids. This suggests that the LiF interlayer not only improves interfacial contact but also mitigates physical intrusion of the ETL into the porous QD network, leading to enhanced film continuity and reduced defect density.

Fig. 2 summarizes the electrical and photoresponse characteristics of devices with and without the LiF interfacial layer. As shown in Fig. 2a, the LiF-modified device exhibits a reduced dark current of 173 nA/cm² at -0.1 V, nearly one order of magnitude lower than the controlled device.

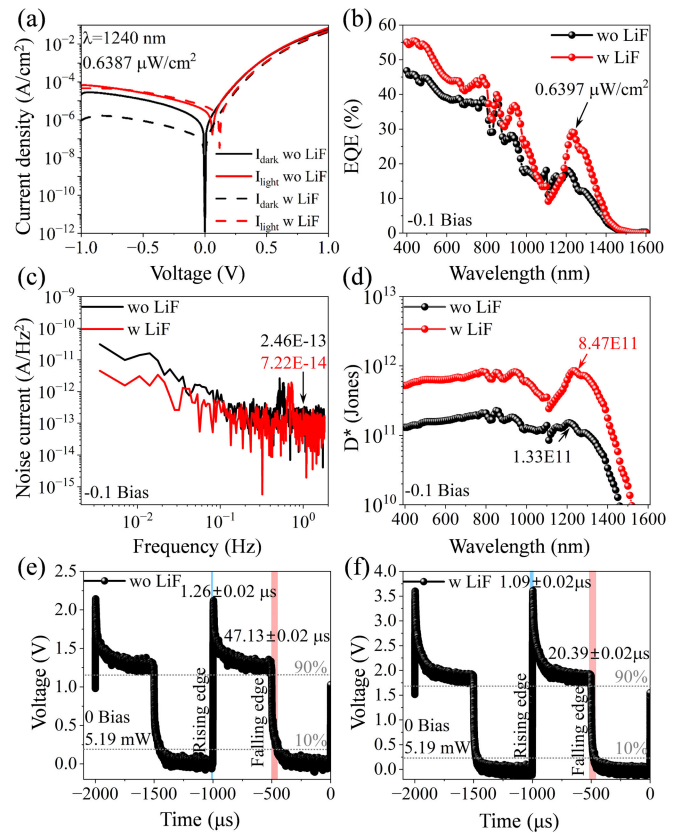


Fig. 2. Device performance comparison with and without LiF interfacial layer. (a) J-V curves under dark and illumination at 1240 nm. (b) EQE spectra. (c) Noise current spectra. (d) Specific detectivity (D^*) spectra. Transient response of devices with (e) and without (f) LiF.

This significant suppression is attributed to the passivation of interfacial traps by the LiF layer, which improves film uniformity and blocks direct contact between the PbS ink and the overlying ETL.

The external quantum efficiency (EQE) (Fig. 2b) further reflects enhanced charge extraction and reduced recombination. At 1240 nm and -0.1 V bias, the EQE increases from 15.45% to 28.90% (corresponding to the responsivity increases from 0.26 A/W to 0.29 A/W). This enhancement of EQE supports the energy level alignment role of LiF and its ability to minimize carrier loss at the interface. Noise current spectra (Fig. 2c) indicate reduced low-frequency noise in the LiF device, due to decreased trap-assisted fluctuations. Consequently, as shown in Fig. 2d, the specific detectivity (D^*) exceeds 8.47×10^{11} Jones, representing more than 6-fold improvement. These results collectively confirm that the ultrathin LiF layer effectively enhances both photon generated charge extraction and noise current suppression, validating its role as a functional interface modifier in inverted PbS QD photodetectors. As shown in the transient response curves (Figs. 2e, f), both devices exhibit an initial overshoot followed by a drop in steady-state photocurrent, indicative of parasitic capacitance effects. Notably, the LiF device exhibits a faster response speed ($1.09 \mu\text{s}$ vs. $1.26 \mu\text{s}$), reflecting an effective carrier transport introduced by the additional interface layer.

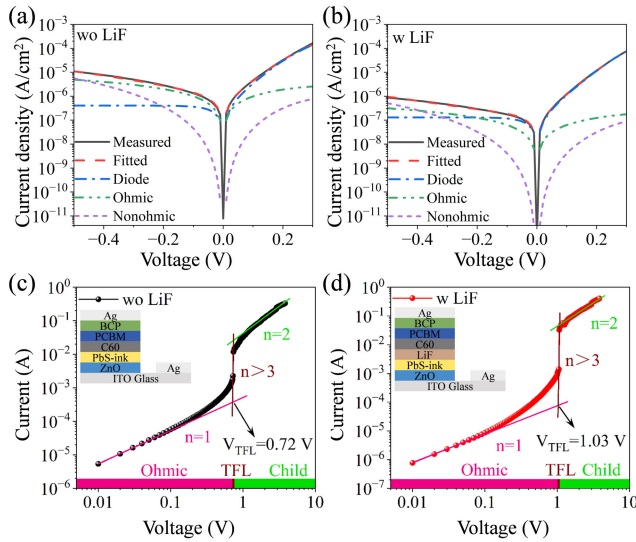


Fig. 3. Dark current component fitting for devices with (a) and without (b) LiF, showing reduced current leakage in the LiF device. SCLC results for devices without (c) and with (d) LiF.

TABLE I
PERFORMANCE COMPARISON OF INVERTED PbS QD PD

Main structure	J_D ($\mu\text{A}/\text{cm}^2$)	D^* (Jones)	Year
NiOx/PbS-BDT/PbS-TBAI/ZnO NPs	0.467 @-0.5V	4.36E10 @-2V	2019 ^[16]
NiOx/PbS/TFCA	7.07 @-0.5V	-	2023 ^[17]
NiOx/PbS-EDT/PbS/C ₆₀ /ZnO	0.05 @-0.5V	-	2023 ^[18]
NiOx/PbS CQD/C ₆₀ /ZnO	0.017 @-0.5V	1.23E10 @0V	2022 ^[5]
PbS-EDT/PbS-I/ZnO/Si	0.174 @-0.5V	1.78E11 @-0.25V	2020 ^[19]
NiOx/PbS-EDT/PbS-ink/LiF/C ₆₀ /PCBM/BCP	0.878 @-0.5v	8.47E11 @-0.1V	This work

To clarify the origins of leakage current, the measured dark J-V curves were fitted using a multi-component current model, including diode, Ohmic, and non-Ohmic terms (Figs. 3a, b). The LiF device shows a significantly reduced Ohmic leakage current (suppressed from 497 nA/cm² to 31.5 nA/cm²) at -0.5 V, consistent with suppressed direct tunneling from the PbS ink layer. This aligns with SEM observations that show fewer cracks and improved morphology in LiF-modified devices. Moreover, The LiF device exhibits suppressed non-Ohmic leakage current from 563 nA/cm² to 51.2 nA/cm², which is associated with interface states and residual parasitic capacitance, suggesting modified energy traps and efficient charge transport.

To further analyze the charge transport properties and trap-related leakage in our devices, space-charge-limited current (SCLC) measurements were performed under dark conditions, as shown in Fig. 3c and 3d. The device structure for SCLC analysis is ITO/ZnO/PbS ink/C₆₀/PCBM/BCP/Ag, with or without a LiF interlayer (insets of Fig. 3c and 3d). The trap-filled limit voltage (V_{TFL}) increases from 0.72 V (without LiF)

to 1.03 V (with LiF), suggesting deeper trap occupation or reduced shallow traps at the interface, which correlates with a slight decrease in carrier mobility.

We summarized the specific performance parameters in Table I within the broader landscape of recent research on quantum dot infrared photodetectors with inverted structure. Benefited from balanced dark current suppression and photon generated carrier extraction by the bilayer LiF/C₆₀ interfacial strategy, our device shows the highest D^* over 8.47×10^{11} Jones.

IV. CONCLUSION

Inverted-structure PbS quantum dot photodetectors have garnered significant attention due to their compatibility with CMOS processes. However, conventional heterojunction interfaces suffer from poor contact and increased trap density. A dual-layer interface engineering strategy employing LiF/C₆₀ was adopted: the LiF layer (1 nm) passivates surface traps and improves film flatness, while the C₆₀ layer (10 nm) ensures energy level alignment and electron extraction. This strategy provides a CMOS-compatible interface design approach for high-performance inverted PbS quantum dot short-wave infrared photodetectors, offering significant application value in infrared imaging, communications, and related fields.

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